

Answers: Sections 1.4 and 1.5

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September 30, 2026

Section 1.4

Problem 1

Given

$$A = \{1, 3, 12, 35\}, \quad B = \{3, 7, 12, 20\}$$

and C is the set of prime numbers.

(a) $A \cap B$

The elements common to both A and B are 3 and 12.

$$A \cap B = \{3, 12\}$$

(b) $(A \cup B) \setminus C$

First, $A \cup B = \{1, 3, 7, 12, 20, 35\}$. The prime numbers in this set are 3 and 7. Removing them gives

$$(A \cup B) \setminus C = \{1, 12, 20, 35\}$$

(c) $A \cup (B \setminus C)$

Since $B \setminus C = \{12, 20\}$, we get

$$A \cup (B \setminus C) = \{1, 3, 12, 20, 35\}$$

Remark: no two of the three resulting sets are disjoint (each pair shares the element 12). However, the set in (a) and the set in (b) are each a subset of the set in (c).

Problem 8

(a) Show that $(A \setminus B) \cap C = (A \cap C) \setminus B$.

Let x be any element.

$$\begin{aligned} x \in (A \setminus B) \cap C &\iff (x \in A \wedge x \notin B) \wedge x \in C \\ &\iff (x \in A \wedge x \in C) \wedge x \notin B \\ &\iff x \in (A \cap C) \setminus B. \end{aligned}$$

Therefore,

$$(A \setminus B) \cap C = (A \cap C) \setminus B$$

(b) Show that $(A \cap B) \setminus B = \emptyset$.

If $x \in (A \cap B) \setminus B$, then $x \in B$ and $x \notin B$, which is impossible. Therefore,

$$(A \cap B) \setminus B = \emptyset$$

(c) Show that $A \setminus (A \setminus B) = A \cap B$.

Let x be any element.

$$\begin{aligned} x \in A \setminus (A \setminus B) &\iff x \in A \wedge x \notin (A \setminus B) \\ &\iff x \in A \wedge \neg(x \in A \wedge x \notin B) \\ &\iff x \in A \wedge (x \notin A \vee x \in B) \\ &\iff x \in A \wedge x \in B \\ &\iff x \in A \cap B. \end{aligned}$$

Therefore,

$$A \setminus (A \setminus B) = A \cap B$$

Problem 14

Show that $A \Delta (B \Delta C) = (A \Delta B) \Delta C$.

Use a truth table (T means the element is in the set):

A	B	C	$A \Delta (B \Delta C)$	$(A \Delta B) \Delta C$
F	F	F	F	F
F	F	T	T	T
F	T	F	T	T
F	T	T	F	F
T	F	F	T	T
T	F	T	F	F
T	T	F	F	F
T	T	T	T	T

The two columns are identical for every combination of A, B, C . Therefore,

$$A \Delta (B \Delta C) = (A \Delta B) \Delta C$$

Section 1.5

Problem 2

(a) Let P = Mary will sell her house, Q = Mary can get a good price, R = Mary can find a nice apartment. “Only if” indicates an implication:

$$P \rightarrow (Q \wedge R)$$

(b) Let P = having a good credit history, Q = having an adequate down payment, R = getting a mortgage. Since $P \wedge Q$ is a necessary condition for R ,

$$\boxed{R \rightarrow (P \wedge Q)}$$

(c) Let P = John will drop out of school, Q = someone stops John. “ P unless Q ” means “if not Q , then P .” Therefore,

$$\boxed{\neg Q \rightarrow P}$$

(d) Let P = x is divisible by 4, Q = x is divisible by 6, R = x is prime. Therefore,

$$\boxed{(P \vee Q) \rightarrow \neg R}$$

Problem 8

(a) Show that $(P \rightarrow Q) \wedge (Q \rightarrow R)$ is equivalent to $(P \rightarrow R) \wedge [(P \leftrightarrow Q) \vee (R \leftrightarrow Q)]$.

P	Q	R	LHS	RHS
T	T	T	T	T
T	T	F	F	F
T	F	T	F	F
T	F	F	F	F
F	T	T	T	T
F	T	F	F	F
F	F	T	T	T
F	F	F	T	T

The LHS and RHS columns are identical. Therefore,

$$\boxed{\begin{aligned} &(P \rightarrow Q) \wedge (Q \rightarrow R) \\ &\equiv (P \rightarrow R) \wedge [(P \leftrightarrow Q) \vee (R \leftrightarrow Q)] \end{aligned}}$$

(b) Show that $(P \rightarrow Q) \vee (Q \rightarrow R)$ is a tautology.

P	Q	R	$(P \rightarrow Q) \vee (Q \rightarrow R)$
T	T	T	T
T	T	F	T
T	F	T	T
T	F	F	T
F	T	T	T
F	T	F	T
F	F	T	T
F	F	F	T

The final column is always true. Therefore,

$$\boxed{(P \rightarrow Q) \vee (Q \rightarrow R) \text{ is a tautology}}$$

Problem 12

Determine which formulas are equivalent:

- (a) $P \rightarrow (Q \rightarrow R)$
- (b) $Q \rightarrow (P \rightarrow R)$
- (c) $(P \rightarrow Q) \wedge (P \rightarrow R)$
- (d) $(P \wedge Q) \rightarrow R$
- (e) $P \rightarrow (Q \wedge R)$

(a)

$$\begin{aligned} P \rightarrow (Q \rightarrow R) &\equiv P \rightarrow (\neg Q \vee R) \\ &\equiv \neg P \vee \neg Q \vee R \end{aligned}$$

(b)

$$\begin{aligned} Q \rightarrow (P \rightarrow R) &\equiv Q \rightarrow (\neg P \vee R) \\ &\equiv \neg Q \vee \neg P \vee R \\ &\equiv \neg P \vee \neg Q \vee R \end{aligned}$$

Thus $\boxed{(a) \equiv (b)}$.

(c)

$$\begin{aligned} (P \rightarrow Q) \wedge (P \rightarrow R) &\equiv (\neg P \vee Q) \wedge (\neg P \vee R) \\ &\equiv \neg P \vee (Q \wedge R) \\ &\equiv P \rightarrow (Q \wedge R) \end{aligned}$$

Thus $\boxed{(c) \equiv (e)}$.

(d)

$$\begin{aligned} (P \wedge Q) \rightarrow R &\equiv \neg(P \wedge Q) \vee R \\ &\equiv \neg P \vee \neg Q \vee R \end{aligned}$$

This is the same as (a) and (b). Therefore $\boxed{(a) \equiv (b) \equiv (d)}$.

Final answer for #12:

$$\boxed{(a) \equiv (b) \equiv (d) \quad \text{and} \quad (c) \equiv (e)}$$